

From (1) we know $\forall x (M(x) \leftrightarrow \neg C(x))$ is True. By defn. of universal quantifier, for all elements b in the class, $M(b) \leftrightarrow \neg C(b)$ is True. $\therefore M(a) \leftrightarrow \neg C(a)$ is True. By defn. of biconditional, $\neg C(a)$ is True. By defn. of negation, $\neg(\neg C(a))$ is False. By double negation law, $C(a)$ is False. By defn. of OR, $S(a) \vee C(a)$ is False. By defn. of universal quantifier, $\forall x (S(x) \vee C(x))$ is False.

e) There is a major such that there is a student in the class in every year of study with that major:
 ~~$\exists x \forall y \exists z \exists w ((M(x) \wedge F(y)) \wedge (x \neq y) \wedge (M(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (M(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (M(w) \wedge Se(w)) \wedge ((C(x) \wedge F(y)) \wedge (x \neq y) \wedge (C(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (C(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (C(w) \wedge Se(w))))$~~
 $\exists x \exists y \exists z \exists w ((M(x) \wedge F(y)) \wedge (x \neq y) \wedge (M(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (M(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (M(w) \wedge Se(w)) \wedge ((C(x) \wedge F(y)) \wedge (x \neq y) \wedge (C(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (C(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (C(w) \wedge Se(w))))$

From the given constraints, we know that there is no mathematics major who is a senior.

$\therefore \forall x (M(x) \Rightarrow \neg S(x))$ is True. From the conditional-disjunction equivalence, $\forall x (\neg M(x) \vee \neg S(x))$ is True. From the 1st De Morgan's law, $\forall x \neg (M(x) \wedge S(x))$ is True. From De Morgan's law of quantifiers, $\neg \exists x (M(x) \wedge S(x))$ is True. By defn. of negation, $\neg(\neg \exists x (M(x) \wedge S(x)))$ is False. By double negation law, $\exists x (M(x) \wedge S(x))$ is False.

$\therefore$ From the defn. of AND, $\exists x \exists y \exists z \exists w ((M(x) \wedge F(y)) \wedge (x \neq y) \wedge (M(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (M(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (M(w) \wedge Se(w)) \wedge ((C(x) \wedge F(y)) \wedge (x \neq y) \wedge (C(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (C(z) \wedge J(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (C(w) \wedge Se(w))))$ is False.

From the given constraints we know that there is no computer science major who is a freshman.

$\therefore \forall x (C(x) \Rightarrow \neg F(x))$ is True. From the conditional-disjunction equivalence, $\forall x (\neg C(x) \vee \neg F(x))$ is True. From the 1st De Morgan's law, $\forall x \neg (C(x) \wedge F(x))$ is True. From De Morgan's law of quantifiers, $\neg \exists x (C(x) \wedge F(x))$ is True.